

Hence by replacing this term in the previous sum, we get the following result.

Lemma D.2 (Deterministic equivalent of $\bar{\mathbf{Q}}\mathbf{A}\bar{\mathbf{Q}}$). *Let $\mathbf{A} \in \mathbb{R}^{p \times p}$ be any deterministic symmetric semi-definite matrix. We have that:*

$$\bar{\mathbf{Q}}\mathbf{A}\bar{\mathbf{Q}} \leftrightarrow \bar{\bar{\mathbf{Q}}}\mathbf{A}\bar{\bar{\mathbf{Q}}} + \left(\frac{\alpha(1-\pi)}{(1+\delta_S)(1+\bar{\delta})} \right)^2 \frac{1}{\bar{n}} \text{Tr}(\bar{\bar{\mathbf{Q}}}\mathbf{A}\bar{\bar{\mathbf{Q}}})\mathbb{E}[\bar{\mathbf{Q}}^2]$$

In particular, we have that:

$$\bar{\mathbf{Q}}^2 \leftrightarrow \frac{1}{\bar{h}} \bar{\bar{\mathbf{Q}}}^2, \quad \bar{\mathbf{Q}}\boldsymbol{\mu}\boldsymbol{\mu}^\top \bar{\mathbf{Q}} \leftrightarrow \bar{\bar{\mathbf{Q}}}\boldsymbol{\mu}\boldsymbol{\mu}^\top \bar{\bar{\mathbf{Q}}}$$

where:

$$\bar{h} = 1 - \left(\frac{\alpha(1-\pi)}{(1+\delta_S)(1+\bar{\delta})} \right)^2 \frac{1}{\bar{n}} \text{Tr}(\bar{\bar{\mathbf{Q}}}^2)$$

D.3 USEFUL RESULTS:

Here we will list all the results with $\bar{\bar{\mathbf{Q}}}$ that will be useful in this analysis. Let us denote by a, b the following quantities:

$$a = \left(\frac{\pi}{1+\delta} + \frac{\alpha(1-\pi)}{1+\delta_S} \right), \quad b = \gamma + \frac{\pi}{1+\delta} + \frac{\alpha(1-\pi)}{(1+\delta_S)(1+\bar{\delta})}$$

such that:

$$\bar{\bar{\mathbf{Q}}} = (a\boldsymbol{\mu}\boldsymbol{\mu}^\top + b\mathbf{I}_p)^{-1} \quad (28)$$

By Sherman-Morrisson's lemma A.3, we have that:

$$\bar{\bar{\mathbf{Q}}} = \frac{1}{b} \left(\mathbf{I}_p - \frac{a\boldsymbol{\mu}\boldsymbol{\mu}^\top}{b + a\|\boldsymbol{\mu}\|^2} \right), \quad \bar{\bar{\mathbf{Q}}}\boldsymbol{\mu} = \frac{\boldsymbol{\mu}}{b + a\|\boldsymbol{\mu}\|^2} \quad (29)$$

We also have that the constants a_1, a_2, b_1 and b_2 from lemma A.5 become by taking their expectations on $\mathbf{z}$:

Lemma D.3 (New values of constants).

$$a_1 = \frac{\pi}{N(1+\delta)^2} \frac{1}{\bar{h}} \text{Tr}(\bar{\bar{\mathbf{Q}}}^2), \quad b_1 = \frac{\alpha(1-\pi)}{N(1+\delta_S)^2} \frac{1}{\bar{h}(1+\bar{\delta})^2} \text{Tr}(\bar{\bar{\mathbf{Q}}}^2)$$

$$a_2 = \frac{\pi}{N(1+\delta)^2} \frac{1}{\bar{h}(1+\bar{\delta})^2} \text{Tr}(\bar{\bar{\mathbf{Q}}}^2), \quad b_2 = \frac{\alpha(1-\pi)}{N(1+\delta_S)^2} \frac{1}{\bar{h}(1+\bar{\delta})^4} \text{Tr}(\bar{\bar{\mathbf{Q}}}^2)$$

where:

$$\frac{1}{N} \text{Tr}(\bar{\bar{\mathbf{Q}}}^2) = \frac{\eta}{b^2} \quad (30)$$

D.4 TEST EXPECTATION:

It only suffices to apply the expectation on $\mathbf{z}_i$ to m_q obtained with the general model in theorem B.1. Hence:

$$\begin{aligned} \mathbb{E}[\mathbf{w}_q^\top \mathbf{x}] &= (-1)^a \left(\frac{\pi}{1+\delta} + \frac{\lambda(1-\pi)}{1+\delta_S} \right) \mathbb{E}[\boldsymbol{\mu}^\top \bar{\bar{\mathbf{Q}}}\boldsymbol{\mu}] \\ &= (-1)^a \left(\frac{\pi}{1+\delta} + \frac{\lambda(1-\pi)}{1+\delta_S} \right) \boldsymbol{\mu}^\top \bar{\bar{\mathbf{Q}}}\boldsymbol{\mu} \\ &= (-1)^a \left(\frac{\pi}{1+\delta} + \frac{\lambda(1-\pi)}{1+\delta_S} \right) \frac{\|\boldsymbol{\mu}\|^2}{b + a\|\boldsymbol{\mu}\|^2} \end{aligned}$$

D.5 TEST VARIANCE:

Using theorem B.1, we need to apply the expectation on $\mathbf{z}$ to the following second order moment:

$$\begin{aligned} \nu_q = & \left(\frac{\pi}{1+\delta} + \frac{\lambda(1-\pi)}{1+\delta_S} \right)^2 \boldsymbol{\mu}^\top \mathbb{E}[\mathbf{Q}\Sigma\mathbf{Q}]\boldsymbol{\mu} + \frac{\pi T_1}{(1+\delta)^2} \left(1 - \frac{2\pi}{1+\delta} \boldsymbol{\mu}^\top \bar{\mathbf{Q}}\boldsymbol{\mu} - \frac{2\lambda(1-\pi)}{1+\delta_S} \boldsymbol{\mu}^\top \bar{\mathbf{Q}}\boldsymbol{\mu} \right) \\ & + \frac{(1-\pi)T_2}{(1+\delta_S)^2} \left(\alpha - \frac{2\lambda^2(1-\pi)}{1+\delta_S} \boldsymbol{\mu}^\top \bar{\mathbf{Q}}\boldsymbol{\mu} - \frac{2\lambda\pi}{1+\delta} \boldsymbol{\mu}^\top \bar{\mathbf{Q}}\boldsymbol{\mu} \right) \end{aligned}$$

where:

$$T_1 = \frac{1}{N} \text{Tr}(\Sigma \mathbb{E}[\mathbf{Q}\Sigma\mathbf{Q}]), \quad T_2 = \frac{1}{N} \text{Tr}(\Sigma_\beta \mathbb{E}[\mathbf{Q}\Sigma\mathbf{Q}])$$

and these two quantities are obtained using corollary A.6 and lemma D.3, which after simplification are given by:

$$T_1 = \frac{(1+\delta)^2}{\pi h} a_1, \quad T_2 = \frac{(1+\delta_S)^2}{\alpha(1-\pi)h} b_1 \quad (31)$$

Now we should define the new deterministic equivalent of $\bar{\mathbf{Q}}\Sigma\bar{\mathbf{Q}}$ to obtain an expression of $\mathbb{E}_{\mathbf{z}}[\mathbb{E}[\mathbf{Q}\Sigma\mathbf{Q}]]$ and to finish this calculus !

Let :

$$\bar{h} = 1 - \left(\frac{\alpha(1-\pi)}{(1+\delta_S)(1+\bar{\delta})} \right)^2 \frac{1}{\bar{h}} \text{Tr}(\bar{\mathbf{Q}}^2) \quad (32)$$

Then, using lemma D.2 we have that the following identities stand for any linear form:

$$\mathbb{E}[\bar{\mathbf{Q}}\Sigma\bar{\mathbf{Q}}] = \bar{\mathbf{Q}}\boldsymbol{\mu}\boldsymbol{\mu}^\top \bar{\mathbf{Q}} + \frac{1}{\bar{h}} \bar{\mathbf{Q}}^2, \quad \mathbb{E}[\bar{\mathbf{Q}}\Sigma_\beta\bar{\mathbf{Q}}] = \bar{\mathbf{Q}}\boldsymbol{\mu}\boldsymbol{\mu}^\top \bar{\mathbf{Q}} + \frac{1}{(1+\bar{\delta})^2} \frac{1}{\bar{h}} \bar{\mathbf{Q}}^2$$

Thus:

$$\begin{aligned} \mathbb{E}_{\mathbf{z}}[\mathbb{E}[\mathbf{Q}\Sigma\mathbf{Q}]] &= \frac{1-b_2}{h} \mathbb{E}[\bar{\mathbf{Q}}\Sigma\bar{\mathbf{Q}}] + \frac{b_1}{h} \mathbb{E}[\bar{\mathbf{Q}}\Sigma_\beta\bar{\mathbf{Q}}] \\ &= \frac{1}{h} \left((1-b_2) \left(\bar{\mathbf{Q}}\boldsymbol{\mu}\boldsymbol{\mu}^\top \bar{\mathbf{Q}} + \frac{1}{\bar{h}} \bar{\mathbf{Q}}^2 \right) + b_1 \left(\bar{\mathbf{Q}}\boldsymbol{\mu}\boldsymbol{\mu}^\top \bar{\mathbf{Q}} + \frac{1}{\bar{h}(1+\bar{\delta})^2} \bar{\mathbf{Q}}^2 \right) \right) \\ &= \frac{1}{h} \left((1+b_1-b_2) \bar{\mathbf{Q}}\boldsymbol{\mu}\boldsymbol{\mu}^\top \bar{\mathbf{Q}} + \frac{1}{\bar{h}} \left(1-b_2 + \frac{b_1}{(1+\bar{\delta})^2} \right) \bar{\mathbf{Q}}^2 \right) \\ &= \frac{1}{h} \left((1+b_1-b_2) \bar{\mathbf{Q}}\boldsymbol{\mu}\boldsymbol{\mu}^\top \bar{\mathbf{Q}} + \frac{1}{\bar{h}} \bar{\mathbf{Q}}^2 \right) \end{aligned}$$

because:

$$\frac{b_1}{(1+\bar{\delta})^2} = b_2$$

Finally, we get the second order moment:

$$\begin{aligned}
\mathbb{E}[(\mathbf{w}_q^\top \mathbf{x})^2] &= \left(\frac{\pi}{1+\delta} + \frac{\lambda(1-\pi)}{1+\delta_S} \right)^2 \frac{1}{h} \boldsymbol{\mu}^\top \left((1+b_1-b_2) \bar{\bar{\mathbf{Q}}} \boldsymbol{\mu} \boldsymbol{\mu}^\top \bar{\bar{\mathbf{Q}}} + \frac{1}{h} \bar{\bar{\mathbf{Q}}}^2 \right) \boldsymbol{\mu} \\
&+ \frac{\pi T_1}{(1+\delta)^2} \left(1 - \frac{2\pi}{1+\delta} \boldsymbol{\mu}^\top \bar{\bar{\mathbf{Q}}} \boldsymbol{\mu} - \frac{2\lambda(1-\pi)}{1+\delta_S} \boldsymbol{\mu}^\top \bar{\bar{\mathbf{Q}}} \boldsymbol{\mu} \right) \\
&+ \frac{(1-\pi)T_2}{(1+\delta_S)^2} \left(\alpha - \frac{2\lambda^2(1-\pi)}{1+\delta_S} \boldsymbol{\mu}^\top \bar{\bar{\mathbf{Q}}} \boldsymbol{\mu} - \frac{2\lambda\pi}{1+\delta} \boldsymbol{\mu}^\top \bar{\bar{\mathbf{Q}}} \boldsymbol{\mu} \right) \\
&= \frac{1}{h} \left(\frac{\pi}{1+\delta} + \frac{\lambda(1-\pi)}{1+\delta_S} \right)^2 \left((1+b_1-b_2)(\boldsymbol{\mu}^\top \bar{\bar{\mathbf{Q}}} \boldsymbol{\mu})^2 + \frac{1}{h} \boldsymbol{\mu}^\top \bar{\bar{\mathbf{Q}}}^2 \boldsymbol{\mu} \right) \\
&+ \frac{\pi T_1}{(1+\delta)^2} \left(1 - \frac{2\pi}{1+\delta} \boldsymbol{\mu}^\top \bar{\bar{\mathbf{Q}}} \boldsymbol{\mu} - \frac{2\lambda(1-\pi)}{1+\delta_S} \boldsymbol{\mu}^\top \bar{\bar{\mathbf{Q}}} \boldsymbol{\mu} \right) \\
&+ \frac{(1-\pi)T_2}{(1+\delta_S)^2} \left(\alpha - \frac{2\lambda^2(1-\pi)}{1+\delta_S} \boldsymbol{\mu}^\top \bar{\bar{\mathbf{Q}}} \boldsymbol{\mu} - \frac{2\lambda\pi}{1+\delta} \boldsymbol{\mu}^\top \bar{\bar{\mathbf{Q}}} \boldsymbol{\mu} \right) \\
&= \frac{1}{h} c^2 \left((1+b_1-b_2)(\boldsymbol{\mu}^\top \bar{\bar{\mathbf{Q}}} \boldsymbol{\mu})^2 + \frac{1}{h} \boldsymbol{\mu}^\top \bar{\bar{\mathbf{Q}}}^2 \boldsymbol{\mu} \right) \\
&+ \frac{a_1}{h} \left(1 - 2c \boldsymbol{\mu}^\top \bar{\bar{\mathbf{Q}}} \boldsymbol{\mu} \right) + \frac{b_1}{\alpha h} \left(\alpha - 2\lambda c \boldsymbol{\mu}^\top \bar{\bar{\mathbf{Q}}} \boldsymbol{\mu} \right)
\end{aligned}$$

Note that:

$$\boldsymbol{\mu}^\top \bar{\bar{\mathbf{Q}}} \boldsymbol{\mu} = \frac{\|\boldsymbol{\mu}\|^2}{b+a\|\boldsymbol{\mu}\|^2}, \quad \boldsymbol{\mu}^\top \bar{\bar{\mathbf{Q}}}^2 \boldsymbol{\mu} = \frac{\|\boldsymbol{\mu}\|^2}{(b+a\|\boldsymbol{\mu}\|^2)^2}, \quad c = \left(\frac{\pi}{1+\delta} + \frac{\lambda(1-\pi)}{1+\delta_S} \right)$$

Therefore:

$$\mathbb{E}[(\mathbf{w}_q^\top \mathbf{x})^2] = \frac{c\|\boldsymbol{\mu}\|^2}{h(b+a\|\boldsymbol{\mu}\|^2)^2} \left(c(1+b_1-b_2)\|\boldsymbol{\mu}\|^2 + \frac{c}{h} - 2 \left(a_1 + \frac{\lambda b_1}{\alpha} \right) (b+a\|\boldsymbol{\mu}\|^2) \right) + \frac{a_1+b_1}{h}$$

which concludes the proof of the main theorem 4.2 of this paper.